CivicResolve AI

# Municipal Operational Intelligence & Delay Risk Diagnostic

*Generated on **2026-10-24 05:35:25** | Filter Scope: Borough: **ALL** | Category: **ALL** | Window: **2023-01-02 to 2024-12-30***

## 1. Executive Summary

This diagnostic synthesizes **10,000** municipal service requests within the selected filter scope. Of these records, **9,778** have been resolved, and **222** remain active or in-progress. The historical median resolution duration across eligible closed cases is **8.7 hours** (0.4 days). A total of **2,231** cases (22.8%) exceeded the project-defined long-duration threshold of **124.925 hours** (~5.2 days).

| | | | |
|---|---|---|---|
| **Total Intake Records** | 10,000 | **Median Duration (Eligible)** | 8.7h (0.4d) |
| **Closed (Valid Timestamp)** | 9,778 | **Unresolved / Unclosed Backlog** | 222 (active + null date) |
| **Service-Eligible Closed** | 9,777 | **Exceeding 124.9h Benchmark** | 2,231 (22.8%) |

## 2. Top Complaint Categories & Borough Distribution

Historical demand patterns demonstrate pronounced heterogeneity by complaint category and municipal jurisdiction. Top complaint types represent the primary service volume drivers:

| Complaint Category | Volume | Share of Filter | Median Duration |
|---|---|---|---|
| Illegal Parking | 1,466 | 14.7% | 1.3h |
| Noise - Residential | 924 | 9.2% | 0.8h |
| HEAT/HOT WATER | 734 | 7.3% | 37.8h |
| Blocked Driveway | 508 | 5.1% | 1.6h |
| Noise - Street/Sidewalk | 421 | 4.2% | 0.6h |

| Borough | Complaints | Median Duration | Delay Rate (>124.9h) |
|---|---|---|---|
| BROOKLYN | 3,037 | 8.5h | 23.8% |
| QUEENS | 2,370 | 4.9h | 16.0% |
| MANHATTAN | 2,118 | 13.3h | 28.3% |
| BRONX | 2,084 | 14.7h | 24.2% |
| STATEN ISLAND | 383 | 15.3h | 19.1% |

## 3. Leakage-Free Machine Learning Performance

All models were trained exclusively on 2023 intake data ($N=4,711$) using verified intake-time features only. Hyperparameters and preferred pipelines were chosen strictly on 2024-H1 validation data ($N=2,437$). The metrics below reflect the **unbiased final test evaluation** on held-out 2024-H2 records ($N=2,635$):

| Task & Pipeline | Primary Metric | Secondary Metrics | Operational Assessment |
|---|---|---|---|
| **Classification**<br>Logistic Regression (Class-Balanced) | **F1: 0.7093**<br>Recall: 88.7% | Precision: 59.1%<br>ROC-AUC: 0.9262<br>PR-AUC: 0.7227 | Captured 488 of 550 delayed requests; generates 338 false alarms. Suited for screening and review triage. |
| **Regression**<br>Random Forest (Log1p Target) | **MAE: 251.99h**<br>MedAE: 15.59h | RMSE: 947.67h<br>$R^2$: 0.1446 | Typical absolute error is ~15.6 hours; heavy right-skew extreme outliers expand mean absolute error. |

## 4. K-Means Operational Pattern Archetypes

K-Means clustering ($k=5$, Silhouette score: **0.3836**) grouped 274 municipal complaint segments into distinct operational demand archetypes:

| Cluster Archetype | Segments | Avg Median Hours | Delay Rate % | Representative Categories |
|---|---|---|---|---|
| **Cluster 0**: High-Volume Rapid Municipal Dispatch | 14 (5.1%) | 9.4h (0.4d) | 0.2% | Illegal Parking, Noise - Residential, HEAT/HOT WATER |
| **Cluster 1**: Moderate-Volume Routine Maintenance | 129 (47.1%) | 22.2h (0.9d) | 7.5% | Abandoned Vehicle, Dirty Condition, Noise - Commercial |
| **Cluster 2**: Multi-Month Capital & Forestry Programs | 8 (2.9%) | 8903.9h (371.0d) | 100.0% | New Tree Request, Food Establishment, Mobile Food Vendor |
| **Cluster 3**: Backlog-Prone / Specialized Operations | 16 (5.8%) | 517.3h (21.6d) | 53.0% | Damaged Tree, Dead/Dying Tree, Noise - Helicopter |
| **Cluster 4**: Complex Residential Code Enforcement | 107 (39.1%) | 495.8h (20.7d) | 62.0% | UNSANITARY CONDITION, APPLIANCE, DOOR/WINDOW |

## 5. Scientific Limitations & Governance Disclaimers

**1. Statistical Dependence, Not Causation:** High feature importance scores indicate mathematical correlation within historical ticket routing, not physical root causes of delay.
**2. Project-Defined Benchmark vs. Official SLA:** The 124.925-hour cutoff is an empirical 75th percentile from the 2023 training partition, not a regulatory standard or official agency SLA violation.
**3. Geographic & Operational Specificity:** Models are trained on NYC 311 municipal data. They are not calibrated, validated, or representative for Indian municipal corporations (e.g., PCMC).
**4. Screening Tool Notice:** The classifier exhibits high recall (88.7%) with moderate precision (59.1%). It should be deployed solely for supervisor screening and review prioritization, not automated punitive actions.

# 6. Appendix: Reinforcement Learning & Queue Allocation Simulation

**Simulation Parameters:** Horizon: **10 periods** | Capacity: **5 slots/period** | Arrival Batch: **6 items/period** | Training: **15 episodes** ($\alpha$=0.1, $\gamma$=0.95).
The multi-queue simulator evaluated 5 policies across identical complaint arrival streams over 5 random seeds:

| Policy | Mean Reward | Slot Utilization | Items Handled | Mean Wait | Max Starve | True Long Capture |
|---|---|---|---|---|---|---|
| **Q-Learning** | 36.81 ± 5.01 | 100.0% | 50 | 0.75p | 2.8p | **92.7%** |
| **FIFO** | 35.85 ± 5.8 | 100.0% | 50 | 0.82p | 2.0p | **87.3%** |
| **Risk-First** | 36.61 ± 5.25 | 100.0% | 50 | 0.42p | 6.2p | **100.0%** |
| **Risk-Plus-Wait** | 37.23 ± 6.05 | 100.0% | 50 | 0.81p | 2.4p | **85.9%** |
| **Round-Robin** | 28.53 ± 5.68 | 100.0% | 50 | 0.7p | 1.0p | **69.9%** |

*Disclaimer: Simulated waiting periods and retrospective outcome captures demonstrate policy focus under synthetic resource constraints. They do not prove that municipal resolution times would have been causally reduced in live operations.*